

DELHI TECHNOLOGICAL UNIVERSITY



Dept. of Environmental Engineering

REPORT ON Water Leakage System In DTU Campus: Assesment, Causes and Solutions

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Water Leakage System in DTU Campus: Assessment, Causes, and Solutions

1. Introduction

Water is one of the most valuable resources for sustaining life and development. Universities and educational campuses, being microcosms of urban infrastructure, consume a significant amount of water daily for various purposes drinking, sanitation, laboratories, and landscaping.

Delhi Technological University (DTU), spread over 164 acres, serves thousands of students, faculty members, and staff. Over the years, instances of water leakage and wastage have been observed across the campus, especially around hostels, canteens, gardens, and washrooms.

This report aims to study the **current water leakage system**, identify the **major causes of leakage**, assess its **impact on resources**, and propose **sustainable solutions** for water management within the DTU campus.

2. Objectives of the Study

- To identify key areas in DTU campus affected by water leakage.
- To analyze the technical and infrastructural causes behind water wastage.
- To quantify water loss and its potential financial/environmental cost.
- To propose sustainable strategies for efficient water management.
- To raise awareness among students and staff about responsible water use.

3. Methodology

The report is based on field observation, interviews, data collection, and secondary research.

3.1 Data Collection

- **Primary Data:**
 - Surveys and direct observation of hostels, academic blocks, and gardens.
 - Interviews with maintenance staff and administration officials.
- **Secondary Data:**
 - DTU infrastructure records (water consumption reports, pipeline maps).
 - Research papers on water management and leakage detection systems.

3.2 Tools Used

- **Flow measurement tools** (to estimate volume of loss).
- **Questionnaires** (to understand frequency and awareness among users).

4. Observations and Findings

4.1 Identified Leakage Zones

1. **Old Boys Hostel (OBH)** – leaking taps and underground pipeline seepage.
2. **Mechanical Block** – joint pipe leak near the cooling plant.
3. **Canteen Area** – frequent overflow from water storage tanks.
4. **Girls Hostel (GH-2)** – dripping taps and faulty flush systems.
5. **Garden near Admin Block** – damaged sprinkler pipes.

4.2 Quantitative Data

Area	Avg. Leakage (Litres/day)	Estimated Monthly Loss (₹)
OBH	600	₹900
GH-2	450	₹700
Admin Block	350	₹500
Gardens	1000	₹1500
Total	2400 L/day	₹3,600 per month



FIG1. Near Environmental Department

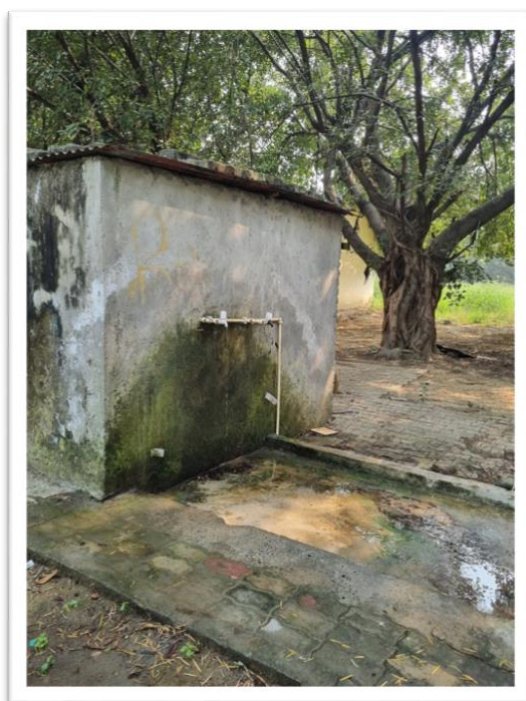


FIG2. Near Sports Complex

5. Proposed Solutions

5.1 Technological Solutions

1. **IoT-Based Smart Water Meters**
 - Install smart sensors to detect flow irregularities in pipelines.
 - Automatic alerts to maintenance team when leakage is detected.
2. **GIS-Integrated Monitoring System**
 - Map all water pipelines digitally.
 - Use satellite data to locate underground seepage.
3. **Automatic Water Shutoff Valves**
 - Cut off water flow during leak detection to prevent wastage.

5.2 Infrastructure Upgradation

- Replace rusted GI pipes with **PVC or HDPE pipelines**.
- Regular inspection every 3 months.
- Proper insulation of underground pipes.

5.3 Management & Awareness

- Conduct workshops on **“Water Conservation at DTU.”**
- Introduce a **‘Leak Report’ mobile app** for students to report issues.
- Display posters near water coolers and hostels about saving water.

6. Cost-Benefit Analysis

Initiative	Estimated Cost	Annual Savings	Payback Period
Smart Sensors Installation	₹1,00,000	₹40,000	2.5 years
Pipe Replacement	₹50,000	₹20,000	2.5 years
Awareness Campaigns	₹10,000	₹5,000	2 years

7. Conclusion

The study highlights that **water leakage in DTU campus** is a manageable yet critical issue. Through systematic assessment and the adoption of modern monitoring techniques, DTU can become a model of water-efficient educational campuses in India.

Reducing leakage not only saves money but also reinforces environmental responsibility aligning with DTU's vision for a **sustainable, smart, and green campus**.

8. References

1. Delhi Jal Board – Water Management Reports, 2023.
2. “Smart Water Leakage Detection Systems Using IoT,” IEEE Paper, 2022.
3. DTU Infrastructure and Maintenance Division Records, 2024.
4. Jal Shakti Abhiyan – Government of India, 2023.